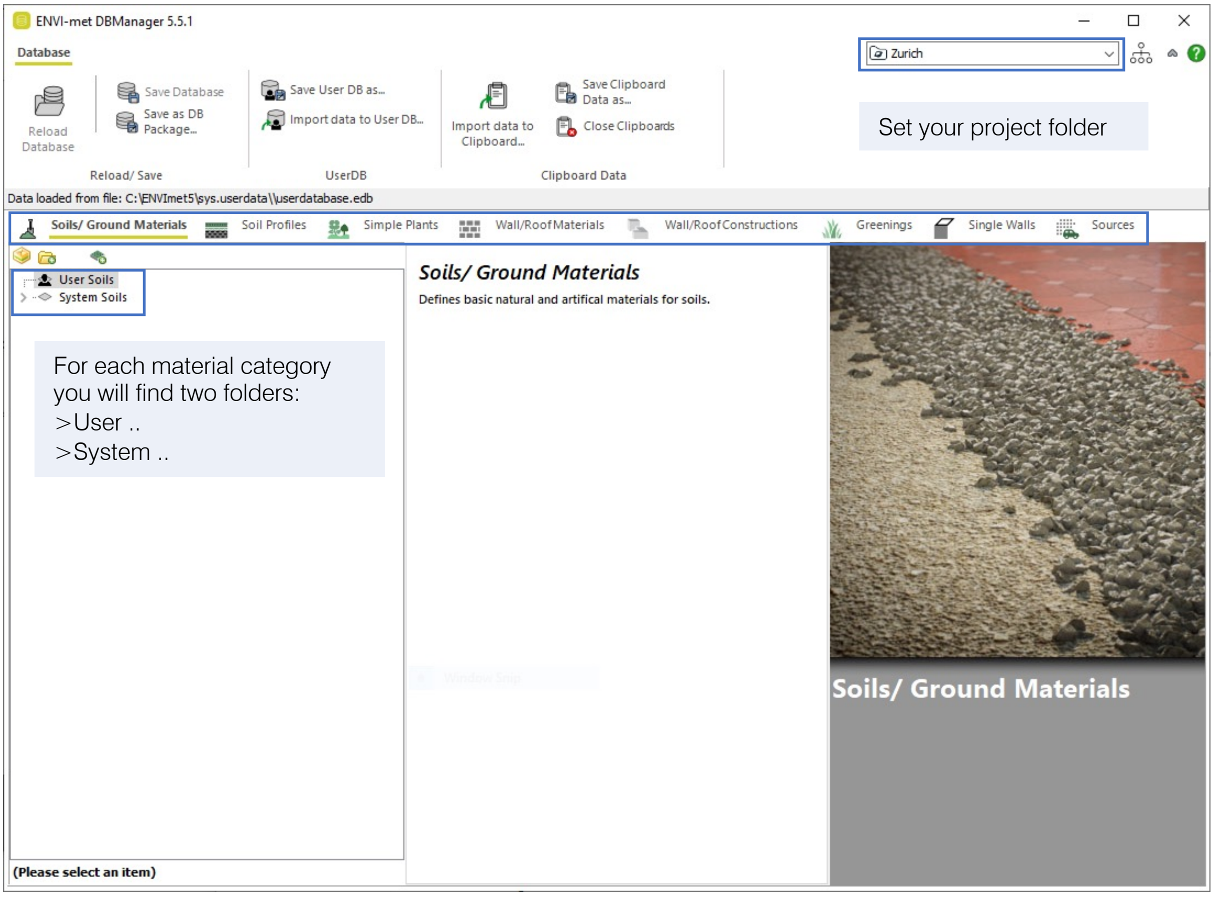
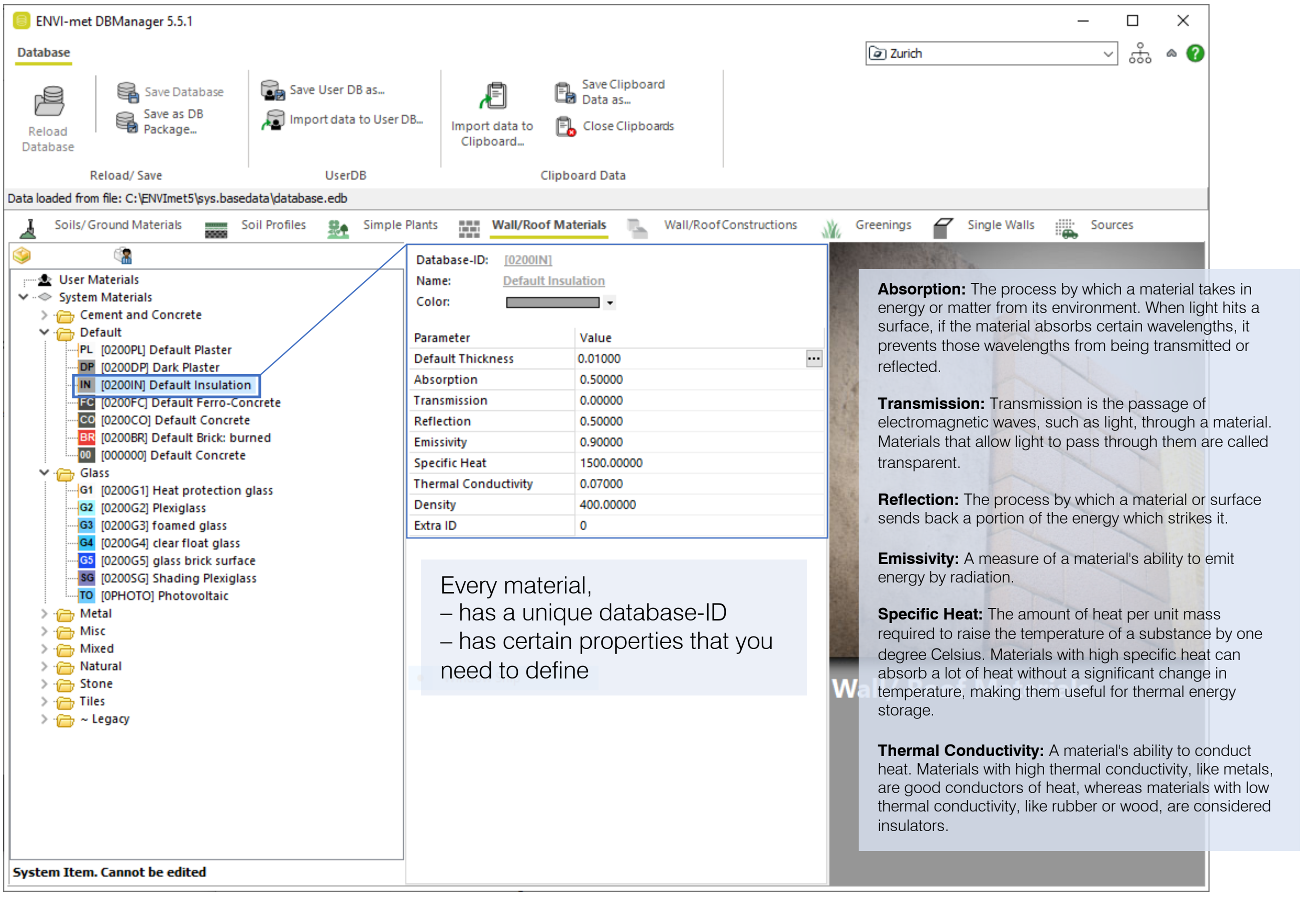
Database

1


ENVI-met DBManager 5.5.1
Database
Reload Database
Save Database
Save as DB Package...
Save User DB as...
Import data to User DB...
Import data to Clipboard...
Save Clipboard Data as...
Close Clipboards
Reload/ Save
UserDB
Clipboard Data
Zurich
Set your project folder
Data loaded from file: C:\ENVImet5\sys.userdata\\userdatabase.edb
Soils/ Ground Materials
Soil Profiles
Simple Plants
Wall/Roof Materials
Wall/Roof Constructions
Greenings
Single Walls
Sources
User Soils
System Soils
For each material category you will find two folders:
>User ..
>System ..
Soils/ Ground Materials
Defines basic natural and artifical materials for soils.
Soils/ Ground Materials
(Please select an item)

2


ENVI-met DBManager 5.5.1
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Save as DB Package...
Save User DB as...
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Import data to Clipboard...
Save Clipboard Data as...
Close Clipboards
Reload/ Save
UserDB
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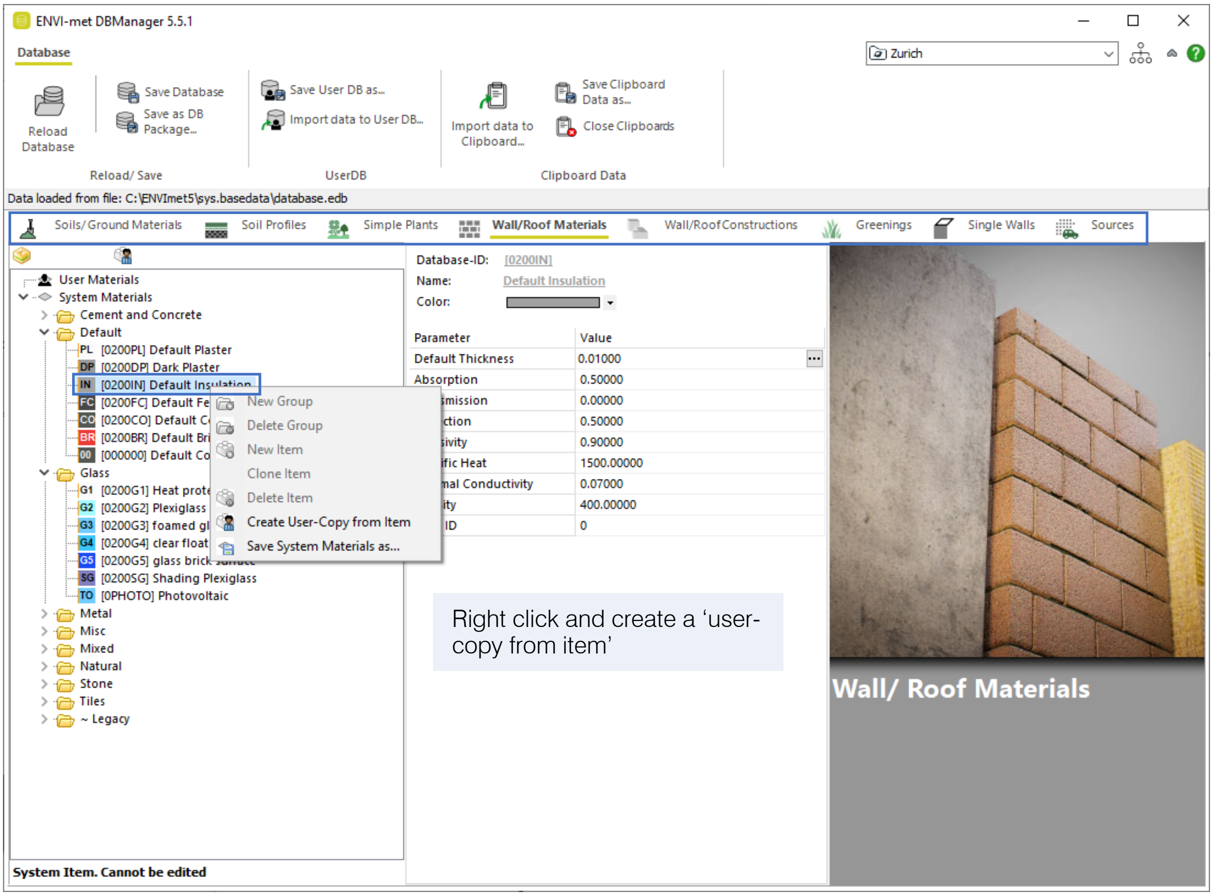
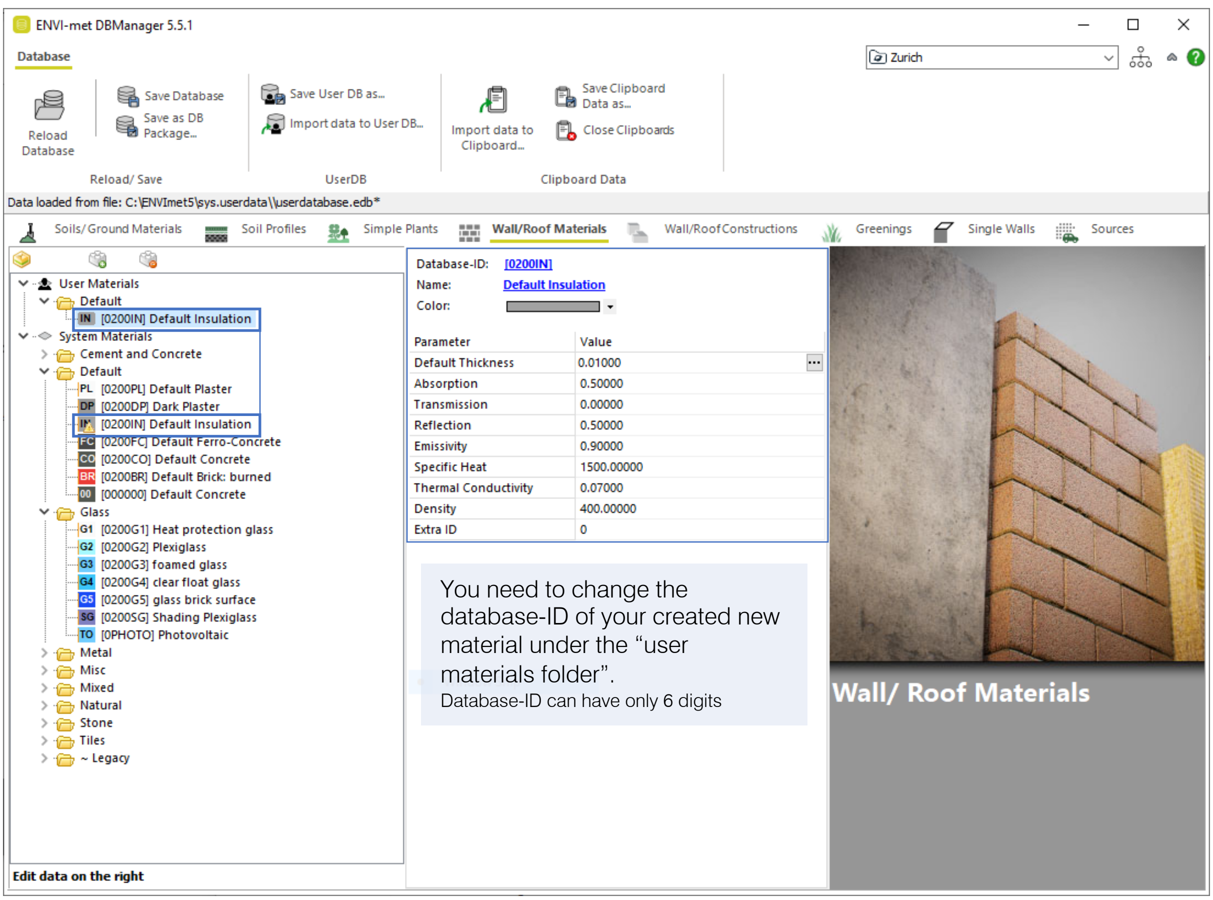
Zurich
Data loaded from file: C:\ENVImet5\sys.basedata\database.edb
Soils/ Ground Materials
Soil Profiles
Simple Plants
Wall/Roof Materials
Wall/Roof Constructions
Greenings
Single Walls
Sources
User Materials
System Materials
Cement and Concrete
Default
[0200PL] Default Plaster
[0200DP] Dark Plaster
[0200IN] Default Insulation
[0200FC] Default Ferro-Concrete
[0200CO] Default Concrete
[0200BR] Default Brick: burned
[000000] Default Concrete
Glass
[0200G1] Heat protection glass
[0200G2] Plexiglass
[0200G3] foamed glass
[0200G4] clear float glass
[0200G5] glass brick surface
[0200SG] Shading Plexiglass
[0PHOTO] Photovoltaic
Metal
Misc
Mixed
Natural
Stone
Tiles
~ Legacy
System Item. Cannot be edited
Database-ID: [0200IN]
Name: Default Insulation
Color:
Parameter | Value
Default Thickness | 0.01000
Absorption | 0.50000
Transmission | 0.00000
Reflection | 0.50000
Emissivity | 0.90000
Specific Heat | 1500.00000
Thermal Conductivity | 0.07000
Density | 400.00000
Extra ID | 0
Every material,
– has a unique database-ID
– has certain properties that you need to define
Absorption: The process by which a material takes in energy or matter from its environment. When light hits a surface, if the material absorbs certain wavelengths, it prevents those wavelengths from being transmitted or reflected.
Transmission: Transmission is the passage of electromagnetic waves, such as light, through a material. Materials that allow light to pass through them are called transparent.
Reflection: The process by which a material or surface sends back a portion of the energy which strikes it.
Emissivity: A measure of a material's ability to emit energy by radiation.
Specific Heat: The amount of heat per unit mass required to raise the temperature of a substance by one degree Celsius. Materials with high specific heat can absorb a lot of heat without a significant change in temperature, making them useful for thermal energy storage.
Thermal Conductivity: A material's ability to conduct heat. Materials with high thermal conductivity, like metals, are good conductors of heat, whereas materials with low thermal conductivity, like rubber or wood, are considered insulators.

Database

3

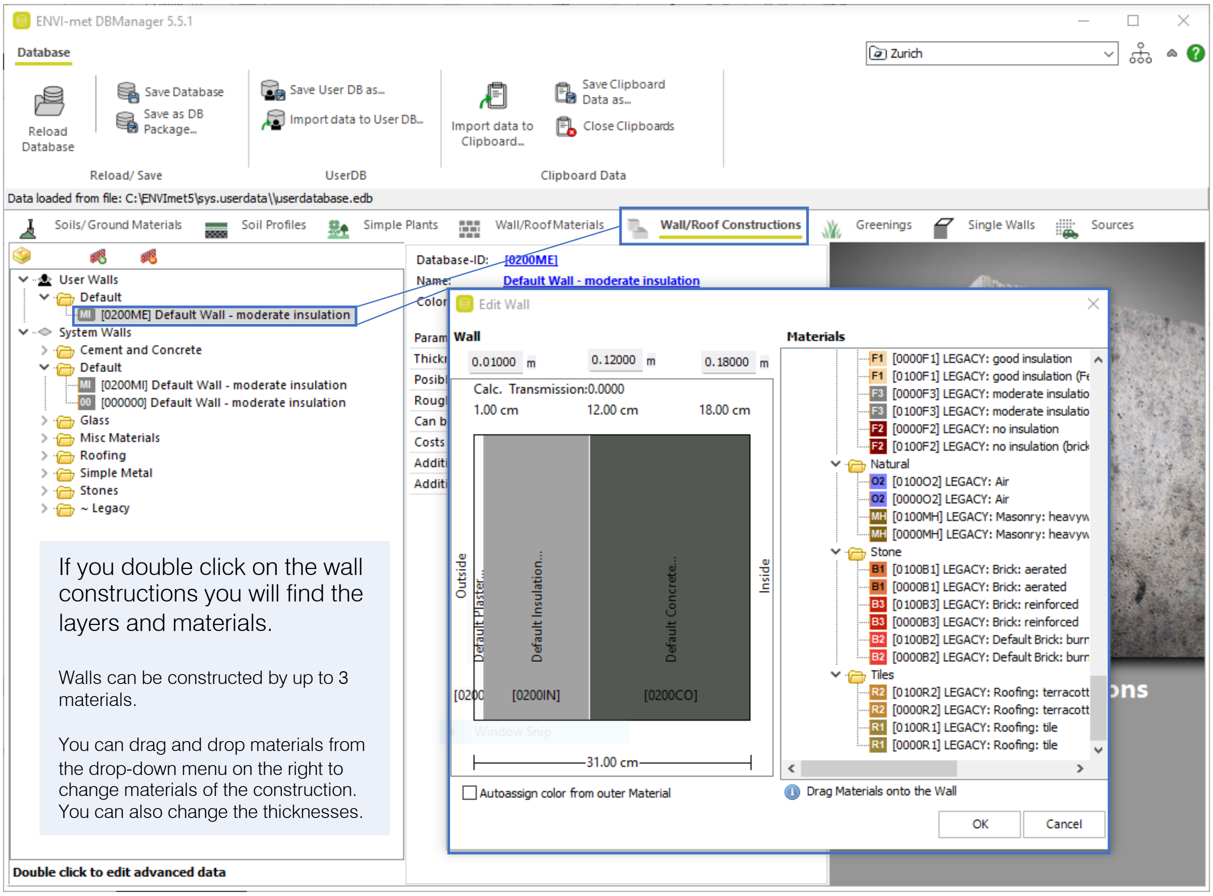

ENVI-met DBManager 5.5.1
Data loaded from file: C:\ENVImet5\sys.basedata\database.edb
New Group
Delete Group
New Item
Clone Item
Delete Item
Create User-Copy from Item
Save System Materials as...
Right click and create a 'user-copy from item'
Wall/ Roof Materials
System Item. Cannot be edited

4


ENVI-met DBManager 5.5.1
Data loaded from file: C:\ENVImet5\sys.userdata\\userdatabase.edb*
You need to change the database-ID of your created new material under the "user materials folder".
Database-ID can have only 6 digits
Wall/ Roof Materials
Edit data on the right

Database

5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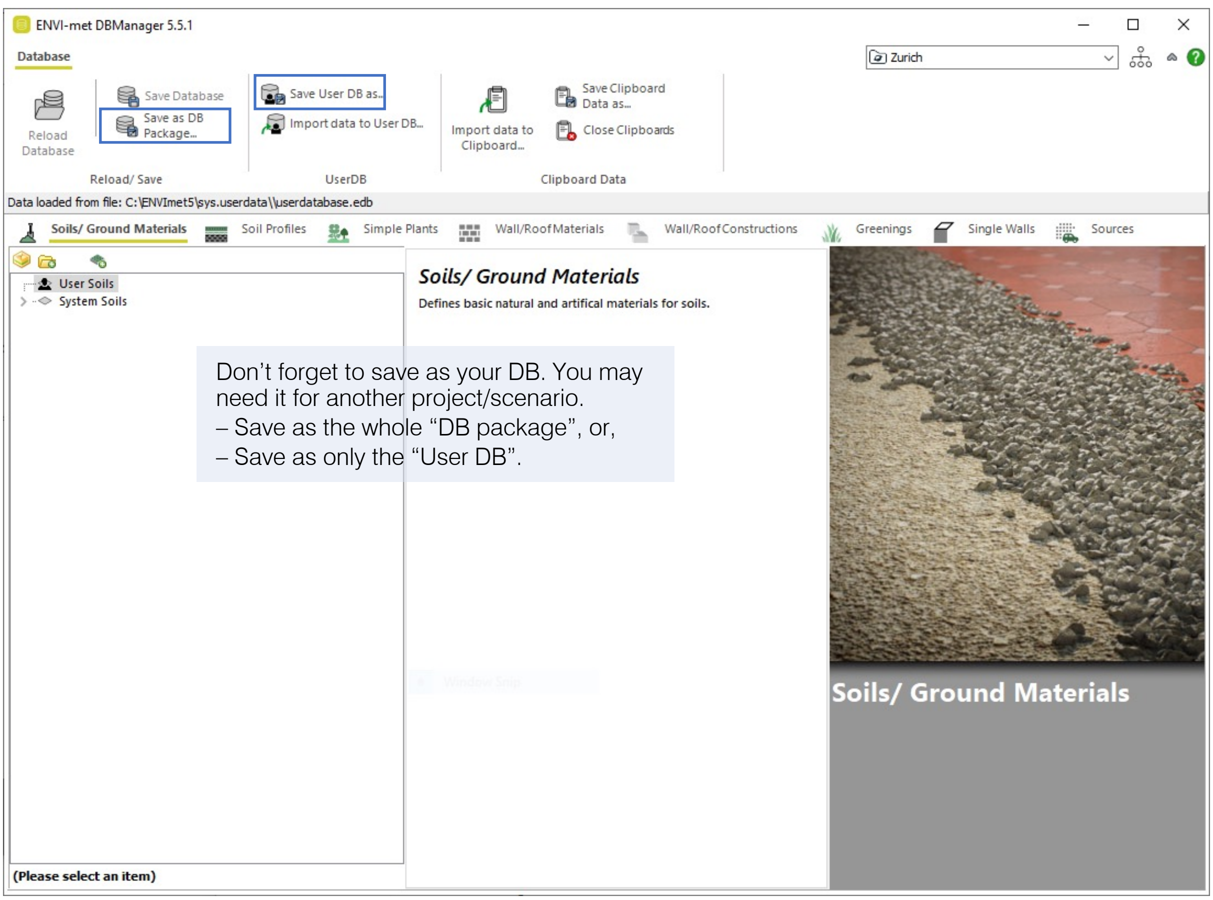
Soils/ Ground Materials
Soil Profiles
Simple Plants
Wall/Roof Materials
Wall/Roof Constructions
Greenings
Single Walls
Sources
User Walls
Default
[0200ME] Default Wall - moderate insulation
System Walls
Cement and Concrete
Default
[0200MI] Default Wall - moderate insulation
[000000] Default Wall - moderate insulation
Glass
Misc Materials
Roofing
Simple Metal
Stones
~ Legacy
If you double click on the wall constructions you will find the layers and materials.
Walls can be constructed by up to 3 materials.
You can drag and drop materials from the drop-down menu on the right to change materials of the construction. You can also change the thicknesses.
Double click to edit advanced data
Database-ID: [0200ME]
Name: Default Wall - moderate insulation
Edit Wall
Wall
0.01000 m
0.12000 m
0.18000 m
Calc. Transmission:0.0000
1.00 cm
12.00 cm
18.00 cm
Outside
Inside
Default Insulation...
Default Concrete...
[0200IN]
[0200CO]
31.00 cm
Autoassign color from outer Material
Materials
[0000F1] LEGACY: good insulation
[0100F3] LEGACY: moderate insulatio
[0000F2] LEGACY: no insulation
Natural
[0100O2] LEGACY: Air
[0000O2] LEGACY: Air
[0100MH] LEGACY: Masonry: heavyw
[0000MH] LEGACY: Masonry: heavyw
Stone
[0100B1] LEGACY: Brick: aerated
[0000B1] LEGACY: Brick: aerated
[0100B3] LEGACY: Brick: reinforced
[0000B3] LEGACY: Brick: reinforced
Tiles
[0100R1] LEGACY: Roofing: tile
[0000R1] LEGACY: Roofing: tile
Drag Materials onto the Wall
OK
Cancel

6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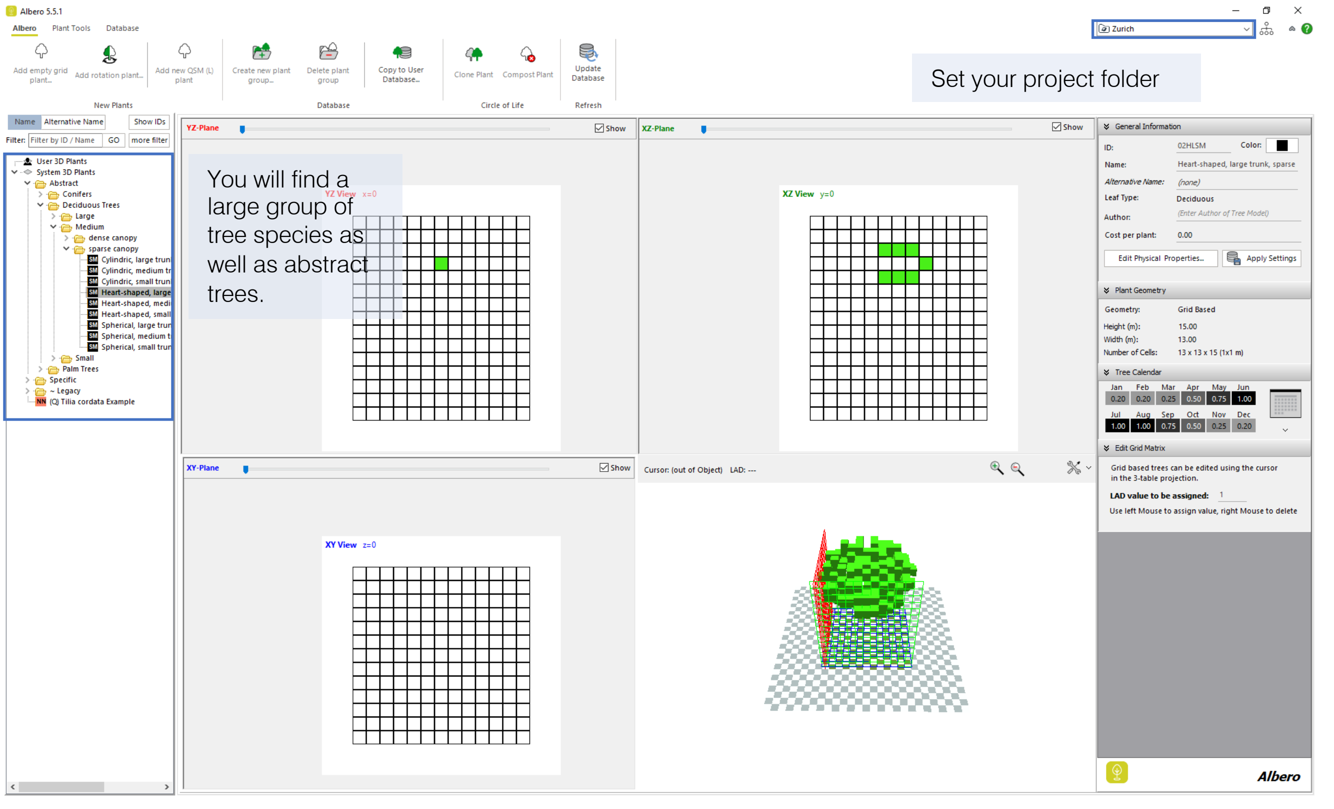
System Soils
Soils/ Ground Materials
Defines basic natural and artifical materials for soils.
Don't forget to save as your DB. You may need it for another project/scenario.
– Save as the whole "DB package", or,
– Save as only the "User DB".
(Please select an item)

Albero

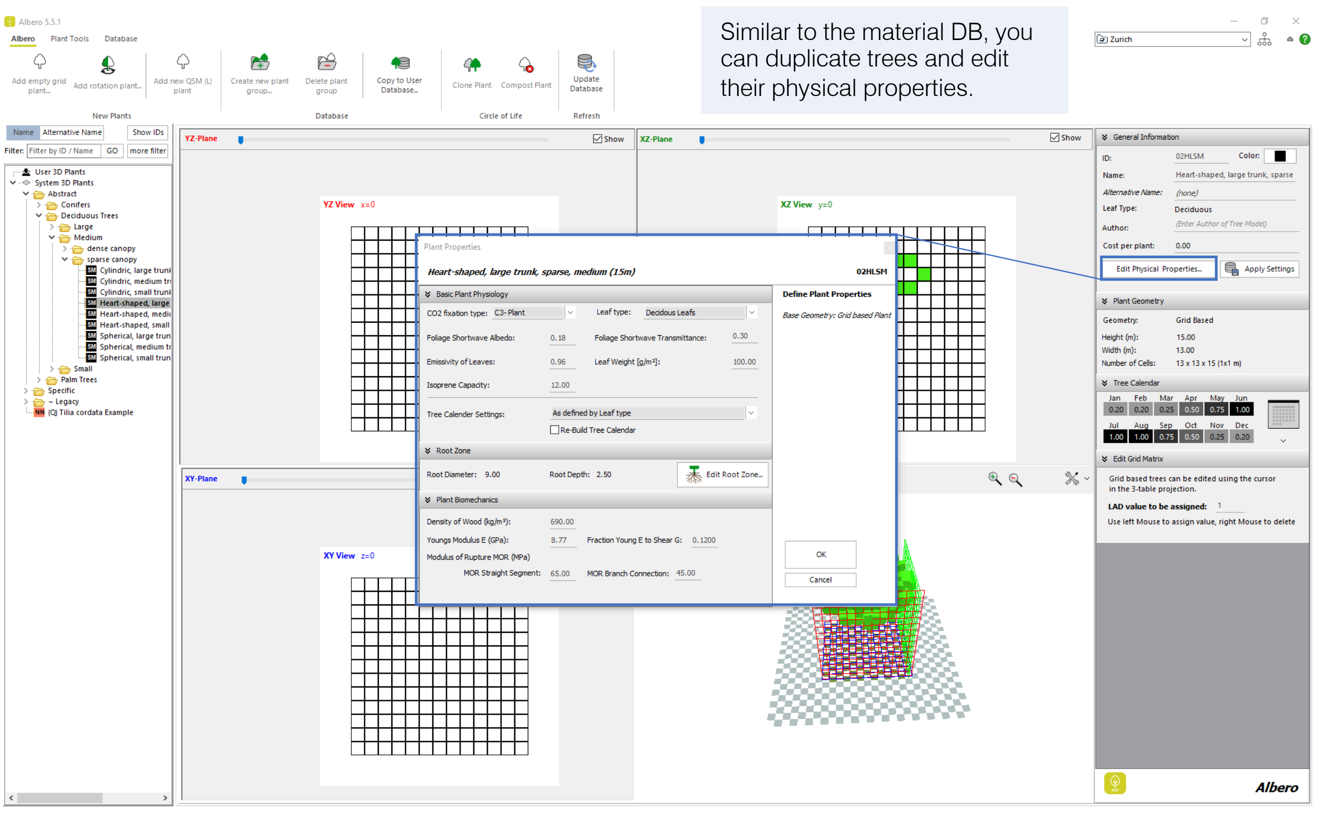
1



Set your project folder

You will find a large group of tree species as well as abstract trees.

2



Similar to the material DB, you can duplicate trees and edit their physical properties.

Albero

3

You can also edit the seasonal leaf factor through “tree calendar”